Client/ Freelancer
:ChatView
:ConversationRoute
:ConversationController
:ConversationsGateway
:ConversationService
:ConversationRepo
:ConversationModel
:DirectMessageModel
:Cloudinary
Database

1. Click the attachment button and select a file
2. handleFileSelect()
3. uploadFile()
4. Forward the multipart request
5. uploadFile(userId, conversationId, file) - the buffer stays in memory

alt **[file == undefined]**
6.1.1. [file == undefined] throw FailedToUploadFileException (500)

[file != undefined]
6.2.1. findConversationByIdForUser(conversationId, userId)
6.2.2. findFirst
6.2.3. Execute query
6.2.4. Return conversation or null
6.2.5. Return conversation or null
6.2.6. Return conversation or null

alt **[conversation == null]**
7.1.1. [conversation == null] throw ConversationNotFoundException (404)

[conversation != null]
7.2.1. uploadConversationFile(file)
7.2.2. Return { secureUrl, bytes }
7.2.3. Return { fileUrl, fileName, fileSize, fileType }
7.2.4. Return 201 - attachment metadata
7.2.5. Return the attachment metadata
7.2.6. sendAttachmentMessage(attachment)
7.2.7. emit('message:send', { conversationId, message: '', attachment })
7.2.8. ensureParticipant(conversationId, senderId)
7.2.9. findConversationByIdForUser(conversationId, senderId)
7.2.10. findFirst
7.2.11. Execute query
7.2.12. Return conversation or null
7.2.13. Return conversation or null
7.2.14. Return conversation or null
7.2.15. Return the conversation (ensureParticipant returns it)
7.2.16. sendMessage(senderId, conversationId, '', attachment)
7.2.17. createMessage(conversationId, senderId, '', attachment)
7.2.18. create
7.2.19. Execute INSERT
7.2.20. Return saved message
7.2.21. Return saved message
7.2.22. Return saved message
7.2.23. Return saved message
7.2.24. server.to(user:{senderId}).to(user:{receiverId}).emit('message:new', { conversationId, message })
7.2.25. emitConversationUpdated() for both rooms - the full flow is in UC-05.5
7.2.26. Return { success: true, message }
7.2.27. Show the image preview or the downloadable file card

alt **[file rejected by conversationFileFilter]**
8.1.1. conversationFileFilter rejects the request - the body never reaches the controller
8.1.2. No attachment is stored and no message is sent

[Cloudinary upload failed]
8.2.1. The Cloudinary upload throws - log the error and throw FailedToUploadFileException
8.2.2. Return the error response
8.2.3. Return the error response
8.2.4. Show "Failed to upload file"